Achievements and failures

Five groups learned to use a two-symbol signal and reached one million simulated years alive in the same run. Two used neural foraging and three used rule-based foraging. Together they finished with 56 living organisms.

What the signal changed

A sender could see which route led to food. A receiver could see only the symbol. Shared group weights started at zero and learned from food feedback. At 750,000 simulated years the weights were frozen, and only then were the control channels shuffled or blocked.

Channel after freezing	Groups alive at 1M	Organisms alive at 1M	Correct food-route choices
Intact learned signal	5 of 6	56	6,656 / 6,656 (100%)
Shuffled	5 of 6	71	6,226 / 12,570 (49.5%)
Blocked, constant symbol	6 of 6	65	6,101 / 12,309 (49.6%)

One neural group assigned symbol 1 to route 0 and symbol 0 to route 1. It chose correctly on 492 and 471 frozen attempts respectively. Both symbols acquired meaning through feedback. The attempt totals include the neural group that died at 986,500 simulated years.

Limit. Better decoding did not improve endpoint survival in the buffered world. The experiment used shared group weights and a supplied channel. It establishes learned signal use at group level, with individual neural learning and inherited communication still untested.

Where communication improved survival

In two separate, tighter tasks, whiteout crossing and ash channels, wrong routes cost energy. All six learned pairs survived across the two tasks and three seeds. All six scripted-perfect pairs also survived. Every shuffled, blocked, and untrained pair died. This is evidence of a survival benefit within those designed tasks.

Eight eras, one continuous run

The schedule spans 2,000 ticks at 500 simulated years per tick. The calendar measures the simulated environmental timeline. Organism ages and generations still advance under the simulator rules.

Simulated years	Era	Main environmental change
0–125,000	Temperate start	Starting ecology
125,000–250,000	Cooling	Lower light and food regrowth
250,000–375,000	Ice age	Lowest light and regrowth in the original schedule
375,000–500,000	Deglaciation	Light recovers and food patches move
500,000–625,000	Warm forest	Higher food regrowth
625,000–750,000	Volcanic disturbance	Temporary hazards on some food-free land
750,000–875,000	Dry age	Food regrowth falls
875,000–1,000,000	Temperate recovery	Light recovers and food patches move again

The original transitions defeated every group

All nine original transition groups entered the ice age, seven reached deglaciation, and three entered the final era. None survived to the million-year endpoint. Static controls finished in five of nine runs.

Foraging and reproduction	Static	Original eras	Refuge eras
Neural, asexual	2 / 3	0 / 3	2 / 3
Neural, social pairing	0 / 3	0 / 3	0 / 3
Rule-based, asexual	3 / 3	0 / 3	3 / 3

The exploratory refuge run added floors for daylight and regrowth and retained half the patches during relocations. Five groups finished. The later communication run used a separate 60% food and regrowth floor. Extinct populations were never replenished.

Two-parent births worked

Neural biparental reproduction occurred in both pairing arms. Each produced one birth, compared with 24 asexual births across the same three seeds. The two pairing births came from the same matched seed.

Normal-resource treatment	Births	Alive at end	Extinctions / 3
Neural asexual	24	0	3
Neural random compatible	1	0	3
Neural social preference	1	0	3
Rule-based asexual	99	33	0
Rule-based random compatible	11	18	0

Biparental births debited 20 energy per parent and gave the child 30 energy. The asexual mode debited 40 energy for the same child endowment. Birth records captured both parents, inherited module sources, and ancestry. Fifteen regression tests passed.

Why the pairing hypothesis remains unresolved

Each neural pairing arm logged 184 mature compatible pair-ticks, with only one pair ready to reproduce. No neural mating offered a choice between two eligible partners, and the 80:20 founder ratio produced no births. At moderate scarcity, no neural pair was ready. The enabled sharing arm recorded only one energy transfer.

Too few neural biparental descendants were born to form matched early and late cohorts. In the supplemental frozen rule-based comparison, descendant survival was identical across inheritance arms within each map. The experiment did not establish a survival advantage from social preference or an inherited improvement in communication.

Evidence files

The v0.3.2 records are in the project experiments directory. The main sources are **signal_ages_v032** (combined runs and cue counts), **signals_v032** (corridor controls), **ages_v032** and **ages_v032_refuges** (era exposure), and **reproduction_v032** (births, ancestry, and transfer exposure).